

Vivekanand Education Society's Institute of Technology, Chembur, Mumbai,
Department Of Computer
Academic Year 2025-26
MID TERM TEST

Q1 a. Enlist the Problems with Traditional Finance

1. High Transaction Fees – Banks and intermediaries charge multiple fees such as service charges, transfer fees, and currency conversion fees.
2. Slow Transactions – International transfers may take 2–5 days due to intermediaries and verification processes.
3. Centralized Control – Financial institutions control transactions and accounts, leading to possible censorship or account freezes.
4. Limited Accessibility – Many people worldwide remain unbanked or underbanked due to strict banking requirements.
5. Lack of Transparency – Users cannot fully verify how institutions process or manage their funds.
6. Intermediaries – Transactions require multiple intermediaries, increasing cost and complexity.
7. Single Point of Failure – Centralized systems can be vulnerable to cyberattacks or system failures.

Q1 b. A person wants to send \$1,000 from the USA to Australia through a traditional bank. The following fees are charged: (1) Exchange rate markup: \$30, (b) Outbound wire fee: \$25, (c) Inbound wire fee: \$15 Determine the amount received in Australia after deducting fees.

Given Fees

- Exchange rate markup = \$30
- Outbound wire fee = \$25
- Inbound wire fee = \$15

Total Fees = 30 + 25 + 15 = \$70

Amount Received = Amount Sent – Total Fees = \$1000 – \$70 = **\$930**

Q1 c. Explain one of the major vulnerability in Decentralized Finance

One major vulnerability in DeFi is Smart Contract Vulnerability.

DeFi applications run on smart contracts, which are self-executing programs deployed on blockchain networks. If there are coding errors or bugs, attackers can exploit them to steal funds.

Ex : In 2021, several DeFi protocols were hacked due to flaws in smart contract logic.

Risks

- Reentrancy attacks
- Logic errors in contract code
- Flash loan exploits
- Unauthorized withdrawals

Since smart contracts are immutable once deployed, fixing vulnerabilities becomes difficult, making security audits essential.

Q1 d. With the help of a neat diagram explain the working of a Liquidity Pool

A Liquidity Pool is a smart contract that holds cryptocurrency tokens and enables trading on decentralized exchanges.

Working

1. Liquidity Providers (LPs) deposit token pairs (e.g., ETH and USDT) into the pool.
2. The pool is managed by a smart contract.
3. Traders swap tokens using the pool instead of a traditional order book.
4. Prices are determined using Automated Market Maker (AMM) algorithms.
5. LPs earn trading fees as rewards for providing liquidity.



Q1 e. Illustrate the key components in Decentralized Finance

1. Blockchain Networks : Provide the infrastructure for decentralized applications.
2. Smart Contracts : Self-executing programs that automate financial operations.
3. Decentralized Applications (DApps) : User interfaces that interact with smart contracts.
4. Liquidity Pools : Pools of tokens used to facilitate trading.
5. Tokens and Cryptocurrencies : Digital assets used for payments, governance, and incentives.
6. Oracles : Provide external real-world data to smart contracts.
7. Wallets : Used by users to store and interact with crypto assets.

Q1 f. What is the role of Oracle in the DeFi Ecosystem?

An **Oracle** is a service that connects **blockchains with external real-world data**.

Blockchains cannot directly access external information such as:

- Market prices
- Weather data
- Sports results
- Interest rates

Role in DeFi

1. Provide **real-time price feeds** for tokens.
2. Enable **accurate lending and borrowing rates**.
3. Support **stablecoins and derivatives**.
4. Trigger smart contract execution based on external events.

Ex : A lending platform needs ETH price data to determine **collateral value**.

Q2 a. Compare Centralized Exchange (CEX) with Decentralized Exchange (DEX)

Feature	Centralized Exchange (CEX)	Decentralized Exchange (DEX)
Control	Controlled by a central authority	No central authority
Custody	Exchange holds user funds	Users keep control of funds
Transaction Speed	Faster	Depends on blockchain speed
Transparency	Limited transparency	Fully transparent
Security	Vulnerable to exchange hacks	Smart contract risks
KYC Requirement	Mandatory	Usually not required
Examples	Coinbase, Binance	Uniswap, SushiSwap

Q2 b. A liquidity pool in a decentralized exchange like Uniswap contains: 50 ETH and 10,000 USDT. Using the AMM rule in the liquidity pool. A trader wants to buy ETH by adding 3,000 USDT to the pool. Determine the new USDT & ETH reserve after the trade. Also find how much ETH the trader receives.

Pool contains: ETH = 50 and USDT = 10,000

Using AMM formula: $x \times y = k$

where : x = ETH reserve and y = USDT reserve

$$k = 50 \times 10,000 = 500,000$$

New USDT reserve = 10,000 + 3,000 = **13,000**

New ETH Reserve : $x \times 13,000 = 500,000$

$$x = \frac{500000}{13000} \approx 38.46 \text{ ETH}$$

ETH Received by Trader = Initial ETH – New ETH = 50 – 38.46 $\approx$ **11.54 ETH**

Q3 a. Describe the concept of Super linear Bonding Curve used for Supply Adjustment in Decentralized Finance with a suitable diagram

A Bonding Curve defines the relationship between token supply and price. In a Superlinear Bonding Curve, price increases faster than supply growth.

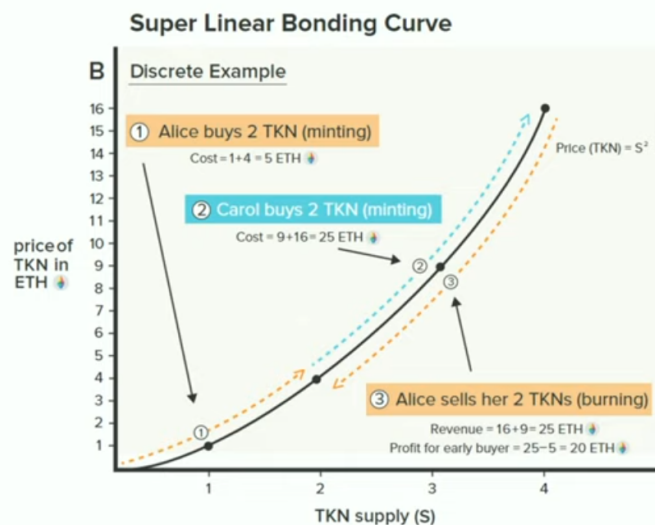
Characteristics

- Price increases rapidly as supply increases
- Encourages early participation
- Prevents excessive token minting

Supply adjustment

Super-linear bonding curves

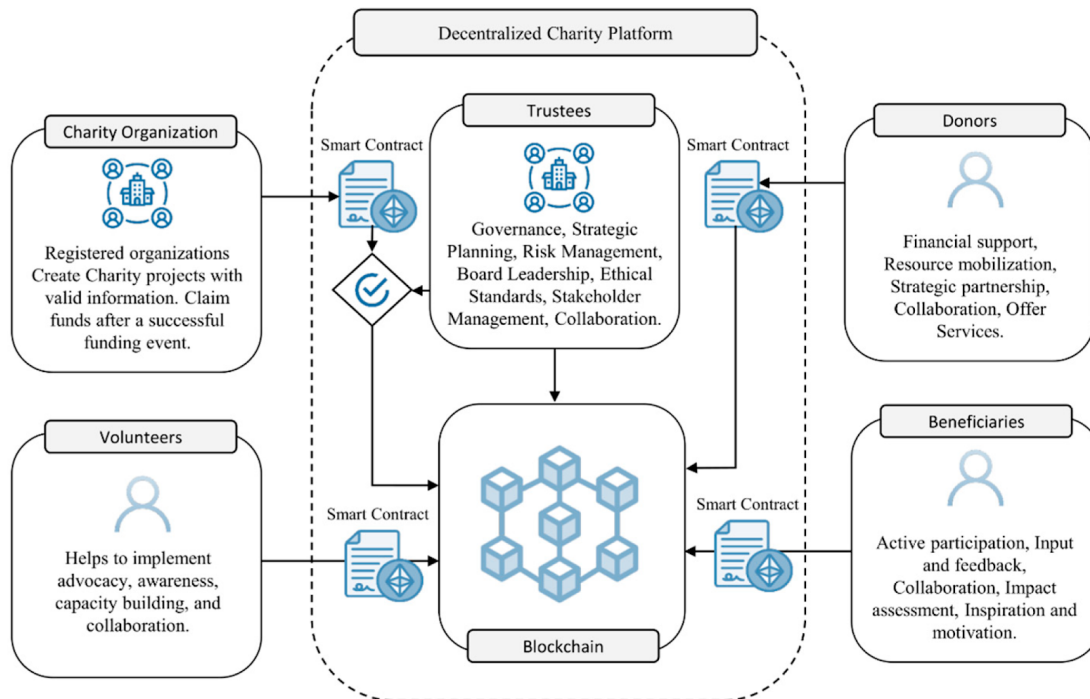
- Example: $TKN = S^2$
- More extreme rewards for early investors



Q3b. With the help of a neat diagram explain the Decentralized Fundraising mechanism.

- Decentralized fundraising allows projects to raise funds directly from investors using blockchain.
- Examples : Initial Coin Offering (ICO) and Initial DEX Offering (IDO)
- To improve transparency, trust, and efficiency in charitable donations.
- Traditional charity systems often suffer from lack of transparency, delayed fund distribution, and misuse of funds.

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- By using blockchain and smart contracts, the platform ensures:
 - Transparent tracking of donations
 - Automated fund distribution
 - Reduced administrative overhead
 - Trust between donors, charities, and beneficiaries
- **5 main stakeholders** connected through smart contracts and managed through the blockchain network.
 1. Charity Organization
 2. Donor
 3. Trustees
 4. Volunteers
 5. Beneficiaries
- **Workflow of the System**
 1. Registration : Charity organizations register and create fundraising campaigns.
 2. Governance Verification : Trustees verify the legitimacy of the organization and project.
 3. Fundraising : Donors send cryptocurrency or digital funds to the campaign through smart contracts.
 4. Transaction Recording : All transactions are stored on the blockchain ledger.
 5. Project Execution : Volunteers help implement the charity project.
 6. Fund Distribution : Smart contracts automatically release funds to charity organizations.
 7. Impact Assessment : Beneficiaries provide feedback and report the impact of the project.

Advantages of the Decentralized Charity Platform

1. Transparency : Every transaction is visible on the blockchain.
2. Reduced Fraud : Smart contracts prevent misuse of funds.
3. Faster Fund Distribution : Funds are transferred automatically without intermediaries.
4. Global Participation : Anyone from anywhere can donate.
5. Accountability : Donors can track the usage of their contributions.